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Hybrid deep learning model for air quality prediction and its impact on healthcare

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Air pollution poses a serious threat to public health, making accurate and timely air quality prediction essential for effective mitigation and planning. This study presents a deep learning–based framework for short-term Air Quality Index (AQI) forecasting that integrates structured feature engineering with advanced neural architectures. Engineered features include lagged pollutant indicators, multi-scale moving averages, seasonal cyclic encodings, pollutant ratios, and date-based temporal variables, designed to capture nonlinear temporal dependencies and pollutant interactions. Air quality data collected from the OpenWeather API are used to evaluate Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and a hybrid CNN–LSTM architecture. Model training is performed using Adam and RMSprop optimizers, and performance is assessed using precision, recall, and F1-score. Experimental results indicate that, within the scope of the evaluated dataset, the hybrid CNN-LSTM model achieves the strongest overall performance for short-term AQI forecasting, attaining an F1-score of approximately 91%, compared with 87.9% for LSTM and 86.7% for CNN under identical configurations. The results further demonstrate that the incorporation of engineered temporal and ratio-based features consistently improves predictive performance across all models. While the study is limited to a region-specific dataset and a short time span, the findings highlight the effectiveness of combining feature engineering with hybrid deep learning architectures for robust AQI prediction and support their potential use in data-driven air quality monitoring systems.

Keywords Air quality index prediction, Deep learning, Hybrid CNN–LSTM, Feature engineering, Temporal modeling, Pollutant interaction, Health risk interpretation

Air pollution refers to the contamination of the atmosphere by harmful substances such as carbon dioxide, nitrogen dioxide, ozone, and particulate matter. Industrial activities, urbanization, and the increasing use of private vehicles are among the primary contributors to deteriorating air quality[38]. Air pollution levels are commonly quantified using the Air Quality Index (AQI), which provides a standardized measure of ambient pollutant concentration and associated health risk.

There are two main categories of air pollutants:

- **Primary Pollutants:** Directly emitted from identifiable sources, including sulfur dioxide (SO_2), carbon monoxide (CO), and particulate matter (PM)[39].
- **Secondary Pollutants:** Formed through chemical reactions involving primary pollutants and atmospheric constituents[40].

Each country faces distinct challenges in air quality monitoring and prediction due to differences in emission sources, meteorological conditions, and urban structure. A major challenge in AQI forecasting arises from sudden changes in atmospheric conditions that alter pollutant dispersion and accumulation[41]. To address this, extensive air quality monitoring networks have been established worldwide to collect high-resolution pollutant data.

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Traditional machine learning approaches, such as Naïve Bayes, K-Nearest Neighbors (KNN), Decision Trees, and ensemble models, have been widely applied for AQI prediction[42]. However, these methods often struggle to capture complex nonlinear relationships and long-range temporal dependencies inherent in air quality data. Consequently, deep learning models have emerged as effective alternatives due to their ability to learn hierarchical representations directly from data.

Among deep learning approaches, Convolutional Neural Networks (CNN) are effective at extracting localized patterns from structured input features, while Long Short-Term Memory (LSTM) networks are well suited for modeling temporal dependencies in sequential data[43]. Rather than reiterating well-established architectural details, this study focuses on leveraging the complementary strengths of these models within a hybrid CNN-LSTM framework for AQI prediction.

Accurate and timely AQI forecasting is essential for pollution mitigation and early warning, particularly in urban environments. Although recent studies have demonstrated promising results using deep learning models, many existing approaches rely on single architectures or emphasize either temporal or spatial dependencies in isolation[44]. Such design choices can limit robustness under rapidly changing atmospheric conditions and heterogeneous pollution patterns. Moreover, performance degradation across regions or temporal scales remains a persistent challenge, indicating that AQI prediction is still an open research problem.

To address these limitations, this study proposes and systematically evaluates a hybrid CNN–LSTM framework augmented with structured and physically interpretable feature engineering. The proposed approach jointly captures local feature representations and long-term temporal dependencies, aiming to improve predictive stability and robustness compared to standalone models.

Impact on health

Air pollution must be controlled at an early stage, and accurate AQI prediction plays a crucial role in this process. Prolonged exposure to polluted air is associated with a wide range of adverse health outcomes, including respiratory illnesses, cardiovascular disease, asthma, and increased mortality risk[45]. Older adults and individuals with pre-existing health conditions are particularly vulnerable.

Air pollution also adversely affects ecosystems, agriculture, and biodiversity. Common mitigation measures include:

- Planting trees and expanding green spaces
- Traffic regulation policies such as odd–even schemes
- Public awareness campaigns
- Expansion of air quality monitoring infrastructure
- Promotion of clean energy alternatives
- Reduction in excessive electricity consumption

Previous studies have established strong links between pollutant exposure and health outcomes. Singh et al. [3] reported that $PM_{2.5}$ directly affects lung function and increases asthma prevalence, while Cincinelli and Martellini [4] showed that indoor air pollution can be equally or more harmful than outdoor exposure. Hedley et al. [5] observed a measurable reduction in respiratory and cardiovascular morbidity following fuel sulfur reduction policies in Hong Kong. Additional studies [6–11] further highlight the socio-economic and regional disparities in pollution exposure and health impact.

The key contributions of this study are summarized as follows:

- A systematic AQI prediction framework is presented that integrates a hybrid CNN–LSTM architecture with explicit and physically interpretable feature engineering, including lagged pollutant indicators, multi-scale moving averages, seasonal cyclic encodings, and pollutant ratio features.
- A comprehensive empirical evaluation of CNN, LSTM, and hybrid CNN–LSTM models is conducted under consistent experimental settings using real-world air quality data, including class-wise performance analysis, confusion matrix interpretation, robustness assessment, and statistical significance testing.
- An interpretable decision-support component is introduced that maps predicted AQI categories to relative health risk indicators, enhancing practical relevance while avoiding epidemiological overinterpretation.

The remainder of this paper is organized as follows. Section 2 reviews related work on air quality forecasting using machine learning and deep learning approaches. Section 3 describes the materials and methods, including dataset description and preprocessing, followed by detailed explanations of the CNN, LSTM, and hybrid CNN–LSTM models. Section 4 outlines the proposed methodology and feature engineering strategy. Section 5 reports the experimental results and provides a detailed discussion of model performance and behavior. Finally, Section 6 concludes the paper and highlights limitations and directions for future work.

Related work

Air quality prediction has been extensively studied using machine learning and deep learning techniques, with a strong emphasis on modeling temporal dependencies and spatio-temporal interactions among pollutants. Early deep learning approaches primarily relied on recurrent architectures. For instance, Ma et al. [12] introduced a TL-BLSTM model that integrates bidirectional LSTM with transfer learning to capture long-term dependencies in $PM_{2.5}$ time series, while Mao et al. [13] proposed the TS-LSTME framework to forecast air quality over a 24-hour horizon using lagged inputs and bi-directional LSTM layers. Although these models demonstrated strong temporal modeling capability, they largely focused on sequence learning without explicitly addressing structured feature engineering or interpretability.

Hybrid learning approaches have subsequently gained attention by combining complementary model architectures. Janarthanan et al. [14] developed an SVR–LSTM hybrid model for AQI classification in the Chennai region, showing that combining statistical learning with recurrent networks can improve predictive accuracy. Similarly, Li et al. [15] employed stacked autoencoders to extract latent spatio-temporal representations for $PM_{2.5}$ prediction. However, these approaches primarily rely on automated feature extraction and provide limited insight into the physical relationships among pollutants.

More recent studies have explored CNN–LSTM hybrid architectures to jointly model spatial and temporal dependencies. Hameed et al. [16] integrated CNN with bi-directional LSTM while incorporating traffic-related factors, and Zhang et al. [17] combined variational mode decomposition with Bi-LSTM to enhance $PM_{2.5}$ forecasting accuracy. Zhao et al. [18] further emphasized the importance of spatio-temporal coupling in deep learning-based air quality prediction. While these hybrid models improve performance, they typically emphasize architectural fusion and often overlook the role of explicitly engineered temporal and ratio-based features.

Comparative studies have consistently shown the superiority of deep learning models over traditional machine learning approaches for AQI prediction. Hunta and Pengchata [19] demonstrated that deep architectures outperform classical models, and Zhan et al. [20] compared GRU, CNN, LSTM, and Bi-LSTM architectures, reporting superior performance for bidirectional recurrent models. Yang et al. [21] highlighted that combining meteorological variables with pollutant concentrations improves predictive accuracy. Nonetheless, these studies focus primarily on performance benchmarking rather than robustness, interpretability, or downstream applicability.

Parallel advances in meteorological forecasting further demonstrate the effectiveness of hybrid deep learning models for nonlinear spatio-temporal data. Gong et al. [31] proposed a CNN–LSTM model for historical temperature prediction, where CNN extracts spatial patterns and LSTM captures temporal dependencies, achieving low MAE and supporting applications in agriculture and energy management. Shen et al. [32] extended this paradigm by introducing a multi-scale CNN–LSTM model with attention mechanisms, achieving an MSE of 1.98 and RMSE of 0.81 for temperature prediction in eastern China, thereby validating the benefits of multi-scale feature fusion and attention-based weighting.

Data decomposition and multi-step forecasting strategies have also been explored to improve prediction stability. Coutinho et al. [33] compared SD-CNN–LSTM and EEMD-CNN–LSTM architectures, showing that while SD-CNN–LSTM performs better for single-step forecasts, EEMD-CNN–LSTM provides superior stability for long-term predictions. Bai et al. [34] proposed a hybrid CNN–BiLSTM model combined with Random Forest for multivariate temperature prediction, achieving MAE and MSE reductions of 35.6% and 57.5%, respectively. These findings highlight the effectiveness of hybridization and decomposition but add significant computational complexity.

In the context of pollutant prediction, Kumar and Kumar [35] introduced an MvS CNN–BiLSTM model that reduced RMSE by 3.8–7.1% for $PM_{2.5}$ prediction. Park et al. [36] employed CNNs with dynamic climate variables for monthly $PM_{2.5}$ forecasting in Seoul, while Yin and Sun [37] combined CEEMD-DWT decomposition with BiLSTM, InceptionV3, and Transformer architectures to reduce MAE by 57.41% in wind power prediction. Although these studies demonstrate strong predictive performance, their complexity and domain-specific design limit general applicability.

In contrast to existing hybrid approaches, the present study emphasizes a unified framework that integrates a hybrid CNN–LSTM architecture with explicit and physically interpretable feature engineering, including lagged pollutant indicators, multi-scale moving averages, seasonal cyclic encodings, and pollutant ratio features. Moreover, the proposed approach extends beyond predictive accuracy by incorporating class-wise evaluation, statistical significance analysis, and an interpretable health-risk mapping component, thereby addressing robustness, interpretability, and applied relevance—dimensions that are often underexplored in prior work.

Materials and methods

Dataset description and preprocessing

The study employs an air quality dataset for **Gurugram, India**, collected from the *Air Pollution API* provided by *OpenWeatherMap*. The dataset comprises **2898 records** spanning the period from **10 January 2024 to 9 May 2024**. The data source is publicly accessible at https://openweathermap.org/api/air-pollution. The OpenWeather Air Pollution API provides historical, current, and forecast air quality data, including hourly measurements of major pollutants such as $PM_{2.5}$, PM_{10}, O_3, NO_2, SO_2, and CO. Access to the dataset requires free user registration and the use of an API key, with standard rate limits imposed by the service provider. All data used in this study were retrieved using publicly available API endpoints under these standard access conditions, ensuring that the dataset can be independently re-collected by other researchers.

The geographic scope of the dataset is restricted to the latitude and longitude coordinates corresponding to the urban region of Gurugram, India. Temporal coverage is limited to the specified four-month interval, reflecting data availability at the time of collection. No proprietary or restricted data sources were used. To ensure high-quality and reliable input for model training, several preprocessing steps were applied. These include timestamp parsing, data cleaning, normalization, and dataset partitioning. Timestamps were standardized into a uniform date-time format to facilitate temporal feature construction and sequential modeling. Records containing missing or invalid pollutant values were removed; the proportion of such records was minimal, and deletion was preferred over imputation to avoid introducing artificial bias into the dataset. Continuous pollutant concentration values were normalized using min–max scaling prior to feature engineering to ensure numerical stability during model training. The processed dataset was subsequently divided into training and testing subsets using a fixed split ratio, with random seed control applied to enable reproducibility of experimental results. The dataset features were normalized using **Z-score normalization** to achieve consistent scaling across variables. The normalization process is expressed as:

$$x_{std} = \frac{x - \mu}{\sigma} \tag{1}$$

where μ and σ denote the mean and standard deviation of each feature, respectively. Subsequently, the data were divided into two subsets: **70% for training** and **30% for testing**, ensuring both adequate learning and fair evaluation.

The predictive model was developed to estimate the **Air Quality Index (AQI)** using major pollutant parameters, namely PM_{10}, $PM_{2.5}$, O_3, SO_2, NO_2, and CO. The AQI was computed as:

$$AQI = \max(SI_{PM10}, SI_{PM2.5}, SI_{O3}, SI_{SO2}, SI_{NO2}, SI_{CO}) \tag{2}$$

where *SI* represents the pollutant-specific sub-index. For instance, the sub-index for PM_{10} is defined as:

$$SI_{PM10} = \frac{PM10}{PM10_{24hrs}} \times 100 \tag{3}$$

Here, $PM10_{24hrs}$ refers to the 24-hour average concentration limit as specified by the *National Ambient Air Quality Standards (NAAQS)*.

To further enhance model performance, several feature engineering techniques were incorporated. These included lag features to capture temporal dependencies, moving averages to smooth short-term fluctuations, ratio features to represent inter-pollutant relationships, time-cyclic encodings to model periodic variations, and month-date features to capture seasonal trends. The integration of these engineered features strengthened the model's ability to learn temporal correlations and improved overall AQI prediction accuracy.

1-D convolutional neural network (1-D CNN)

One-dimensional Convolutional Neural Networks (1-D CNNs) are particularly effective for processing temporal sequences such as pollutant time series for Air Quality Index (AQI) prediction. The 1-D CNN architecture used in this study is composed of the following layers:

- **Input layer:** The dataset contains $N = 2898$ samples with $F = 6$ features (pollutant values). The input is a 1-D array suitable for a 1-D convolutional layer:

$$\mathrm{X} \in \mathbb{R}^{N \times F} \tag{4}$$

- **Convolutional layer:** A total of 64 filters are used, each with a kernel size of 5 and a stride of 1. The convolution operation is defined as:

$$(\mathrm{x} * \mathrm{w})[i] = \sum_{j=0}^{k-1} x[i+j] \cdot w[j] \tag{5}$$

To preserve the input dimension, `same` padding is applied. The ReLU activation function is used:

$$\mathrm{ReLU}(z) = \max(0, z) \tag{6}$$

- **Dropout layer:** A dropout rate of 0.25 is applied to reduce overfitting by randomly deactivating 25% of neurons during training:

$$\mathrm{Dropout}(x) = \begin{cases} 0 & \text{with probability } p = 0.25 \\ x & \text{with probability } 1 - p \end{cases} \tag{7}$$

- **Max pooling layer:** Pooling with a pool size of 2 is performed to reduce the feature map dimensionality and retain the most important features:

$$y[i] = \max(x[2i], x[2i+1]) \tag{8}$$

- **Flatten layer:** All pooled feature maps are converted into a 1-D vector:

$$\mathrm{z} \in \mathbb{R}^{1 \times M} \tag{9}$$

where *M* is the number of flattened units.

- **Dense (hidden) layer:** The flattened vector is passed to a dense hidden layer with 50 units. Weights are initialized using He-uniform initialization, and ReLU is used as the activation:

$$a^{(1)} = \mathrm{ReLU}(W^{(1)} \cdot \mathrm{z} + b^{(1)}) \tag{10}$$

- **Output layer:** For multi-class classification, a softmax activation function is used:

$$\mathrm{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \tag{11}$$

- **Loss function:** The categorical cross-entropy loss function is applied due to the multi-class nature of the classification problem:

$$\mathcal{L} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{12}$$

where C is the number of classes, y_i is the true label, and $\hat{y}_i$ is the predicted probability.

- **Optimizers:** To minimize the loss, Adam and RMSprop optimizers are used.
  - **RMSprop:** Adapts the learning rate using an exponentially weighted average of past squared gradients.
  - **Adam:** Combines RMSprop and momentum:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t \tag{13}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \tag{14}$$

$$\theta_{t+1} = \theta_t - \alpha \cdot \frac{m_t}{\sqrt{v_t} + \epsilon} \tag{15}$$

The model is trained for multiple epochs until the loss function converges to a global minimum. The architecture and layer-wise configuration of the 1-D Convolutional Neural Network used in this study are summarized in Table 1.

Long short-term memory (LSTM)

Long Short-Term Memory (LSTM) networks are a class of Recurrent Neural Networks (RNNs) designed to model sequential data while addressing the vanishing gradient problem. Unlike Convolutional Neural Networks (CNNs), LSTMs incorporate memory cells that allow them to remember previous temporal information, making them well-suited for tasks involving time-series prediction such as AQI estimation.

Each LSTM unit contains an input gate (i_t), forget gate (f_t), output gate (o_t), and a memory cell (c_t). These gates regulate the flow of information as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{16}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{17}$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \tag{18}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{19}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{20}$$

$$h_t = o_t \odot \tanh(c_t) \tag{21}$$

Layer (type)	Output shape	Parameters
`Conv1D (conv1d_8)`	(None, 6, 64)	384
`Conv1D (conv1d_9)`	(None, 6, 64)	20,544
`Dropout (dropout_4)`	(None, 6, 64)	0
`MaxPooling1D (max_pooling1d_4)`	(None, 3, 64)	0
`Flatten (flatten_4)`	(None, 192)	0
`Dense (dense_8)`	(None, 50)	9,650
`Dense (dense_9)`	(None, 4)	204
Total parameters	30,782	
Trainable parameters	30,782	
Non-trainable parameters	0	

Table 1. CNN Summary.

The architecture of the LSTM model in this work consists of the following layers:

- **Input layer:** The input layer has neurons equal to the number of features in the dataset (e.g., pollutant concentrations and engineered variables).
- **LSTM layers:** The model includes three stacked LSTM layers, each with 64 memory units. ReLU activation is applied to the outputs of these layers.
- **Dropout layer:** A dropout rate of 0.25 is used to prevent overfitting by randomly deactivating 25% of neurons during training.
- **Flatten layer:** The output from the final LSTM layer is flattened into a 1-D vector.
- **Hidden dense layer:** A dense layer with 50 units is used, with ReLU activation function and He-uniform weight initialization.
- **Output layer:** The final layer uses the softmax activation function for multi-class classification:

$$\mathrm{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \tag{22}$$

- **Loss function:** The model is trained using the categorical cross-entropy loss function:

$$\mathcal{L} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{23}$$

- **Optimizers:** Both RMSprop and Adam optimizers are employed to minimize the loss. RMSprop adapts the learning rate for each parameter, while Adam incorporates both first and second moments for faster convergence:

$$\theta_{t+1} = \theta_t - \alpha \cdot \frac{m_t}{\sqrt{v_t} + \epsilon} \tag{24}$$

The detailed model summary is presented in Table 2.

Hybrid CNN–LSTM model

The Hybrid model integrates the strengths of both Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks for more effective AQI prediction. While CNN layers perform spatial feature extraction, the LSTM layers are responsible for learning temporal dependencies from the feature maps generated by CNN. This sequential integration enhances both the accuracy and robustness of the model.

The input data is first processed through convolutional and pooling operations to extract local patterns and features. These are then passed into LSTM memory units to capture time-series dependencies before being forwarded to the fully connected layers for final classification.

The architecture of the Hybrid model consists of the following components:

- **Convolutional layers:** Two 1-D convolutional layers are used with 64 filters each and kernel size 5. These layers extract local features from the input sequence:

$$(\mathrm{x} * \mathrm{w})[i] = \sum_{j=0}^{k-1} x[i+j] \cdot w[j] \tag{25}$$

Layer (Type)	Output shape	Parameters
`LSTM (lstm_12)`	(None, 6, 64)	16,896
`LSTM (lstm_13)`	(None, 6, 64)	33,024
`LSTM (lstm_14)`	(None, 64)	33,024
`Dropout (dropout_4)`	(None, 64)	0
`Flatten (flatten_4)`	(None, 64)	0
`Dense (dense_8)`	(None, 50)	3,250
`Dense (dense_9)`	(None, 4)	204
Total parameters	86,398	
Trainable parameters	86,398	
Non-trainable parameters	0	

Table 2. LSTM Model Summary.

- **Dropout and pooling:** Dropout with a rate of 0.25 is applied to prevent overfitting, followed by MaxPooling with pool size 2 to reduce feature dimensionality:

$$y[i] = \max(x[2i], x[2i+1]) \tag{26}$$

- **LSTM layers:** Two LSTM layers are used. The first LSTM layer returns sequences, while the second outputs a fixed-length vector. These layers model temporal relationships:

$$h_t = o_t \odot \tanh(c_t) \tag{27}$$

- **Flatten and dense layers:** The LSTM output is flattened and passed through a dense hidden layer of 50 units (ReLU activation), followed by an output layer with softmax activation:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \tag{28}$$

- **Loss function and optimizers:** Categorical cross-entropy is used for loss calculation:

$$\mathcal{L} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{29}$$

Optimization is done using both RMSprop and Adam algorithms to ensure effective and stable training. Figure 1 illustrates the hybrid model structure. Table 3 presents the detailed layer-wise configuration and parameters.

Methodology

In this study, a deep learning-based framework is proposed for accurate prediction of air quality, incorporating external feature engineering and advanced neural architectures. The methodology involves multiple stages including data acquisition, feature construction, AQI calculation, model training, and health risk estimation—executed in a unified pipeline to enhance predictive accuracy and interpretability. The dataset is collected from the OpenWeather API, which provides real-time and historical air quality data. Six major pollutants are considered in this study: PM_{10}, $PM_{2.5}$, O_3, SO_2, NO_2, and CO. These pollutants are widely regarded as critical indicators for assessing atmospheric pollution. Based on these pollutant concentrations, the Air Quality Index (AQI) is calculated. The AQI is defined as the maximum of the sub-indices of each pollutant. Each sub-index SI_x is computed as:

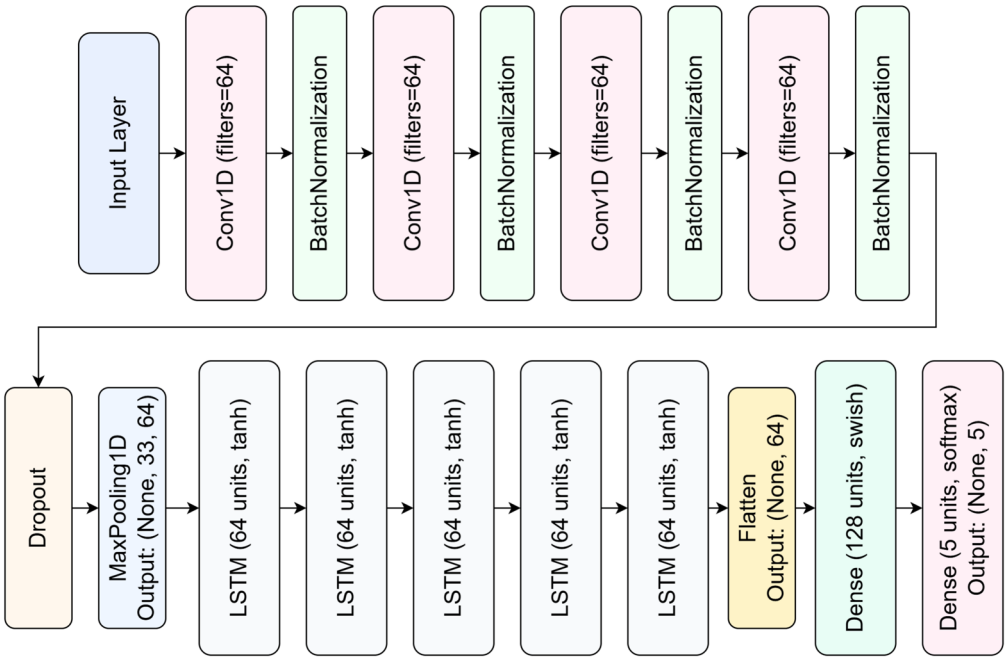

Fig. 1. Hybrid CNN-LSTM Architecture for AQI Prediction.

Layer (type)	Output shape	Parameters
Conv1D (conv1d_8)	(None, 6, 64)	384
Conv1D (conv1d_9)	(None, 6, 64)	20,544
Dropout (dropout_4)	(None, 6, 64)	0
MaxPooling1D (max_pooling1d_4)	(None, 3, 64)	0
LSTM (lstm_8)	(None, 3, 64)	33,024
LSTM (lstm_9)	(None, 64)	33,024
Flatten (flatten_4)	(None, 64)	0
Dense (dense_8)	(None, 50)	3,250
Dense (dense_9)	(None, 4)	204
Total parameters	90,430	
Trainable parameters	90,430	
Non-trainable parameters	0	

Table 3. Hybrid CNN–LSTM Model Summary.

Pollutant	Standard value	Pollutant	Standard value
PM_{10} (24 hrs)	100 $\mu g/m^3$	SO_2 (24 hrs)	80 $\mu g/m^3$
$PM_{2.5}$ (24 hrs)	60 $\mu g/m^3$	NO_2 (24 hrs)	80 $\mu g/m^3$
O_3 (8 hrs)	100 $\mu g/m^3$	CO (8 hrs)	2 mg/m^3

Table 4. Standard Values of Pollutants.

$$SI_x = \left(\frac{x}{x_{\text{standard}}}\right) \times 100 \tag{30}$$

Here, x denotes the observed pollutant concentration and x_{standard} represents the regulatory standard limit for that pollutant. The threshold values for all pollutants, based on established guidelines, are summarized in Table 4.

To enhance the representational capacity of the models, extensive feature engineering is applied to the raw pollutant time series. The engineered features include lagged pollutant values, moving averages, seasonal encodings, pollutant ratios, and temporal indicators, each designed to capture distinct aspects of air quality dynamics.

Lagged features are introduced to explicitly model short-term and delayed temporal dependencies in pollutant behavior. For each pollutant, multiple lag steps are considered, including examples such as $PM_{2.5,t-1}$, $PM_{2.5,t-3}$, $PM_{10,t-1}$, $PM_{10,t-6}$, $NO_{2,t-1}$, and $O_{3,t-3}$, where t denotes the current time index. These features enable the models to learn pollutant persistence effects and delayed responses that frequently occur in atmospheric processes.

Moving average features are computed to reduce short-term noise and emphasize underlying trends. Rolling mean windows of 3-hour, 6-hour, and 12-hour durations are applied to key pollutants such as $PM_{2.5}$, PM_{10}, and NO_2. These aggregated features provide smoothed representations of pollutant evolution and improve robustness against transient measurement fluctuations.

To encode seasonal and periodic variations, cyclic transformations of temporal variables are applied:

$$\text{sin_month} = \sin\left(\frac{2\pi \cdot \text{month}}{12}\right), \quad \text{cos_month} = \cos\left(\frac{2\pi \cdot \text{month}}{12}\right) \tag{31}$$

Additional cyclic encodings are applied to the day-of-week variable using sine and cosine transformations to capture weekly periodicity in emission patterns caused by human activity.

Pollutant ratio features are constructed to model interactions among co-occurring pollutants and their combined influence on air quality. Representative ratios include $PM_{2.5}/PM_{10}$, which reflects the dominance of fine particulate matter, NO_2/O_3, which provides insight into photochemical reactions, and CO/NO_2, which serves as a proxy for traffic-related emissions. These ratios provide complementary contextual information beyond absolute pollutant concentrations and help distinguish between adjacent AQI categories.

Once feature construction is complete, the dataset is split into 70% training and 30% testing data. The target label is the AQI category, derived from the computed AQI values based on predefined thresholds shown in Table 5.

Three deep learning models are implemented for AQI classification: Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), and a hybrid CNN–LSTM architecture. The CNN is employed to extract high-level local patterns from pollutant feature sequences, while the LSTM captures long-term temporal dependencies. The hybrid CNN–LSTM model integrates both architectures by feeding convolutional feature

AQI range	Category	AQI range	Category
0–50	Good	51–100	Satisfactory
101–200	Moderate	201–300	Poor
301–400	Very Poor	>400	Severe

Table 5. AQI Categories.

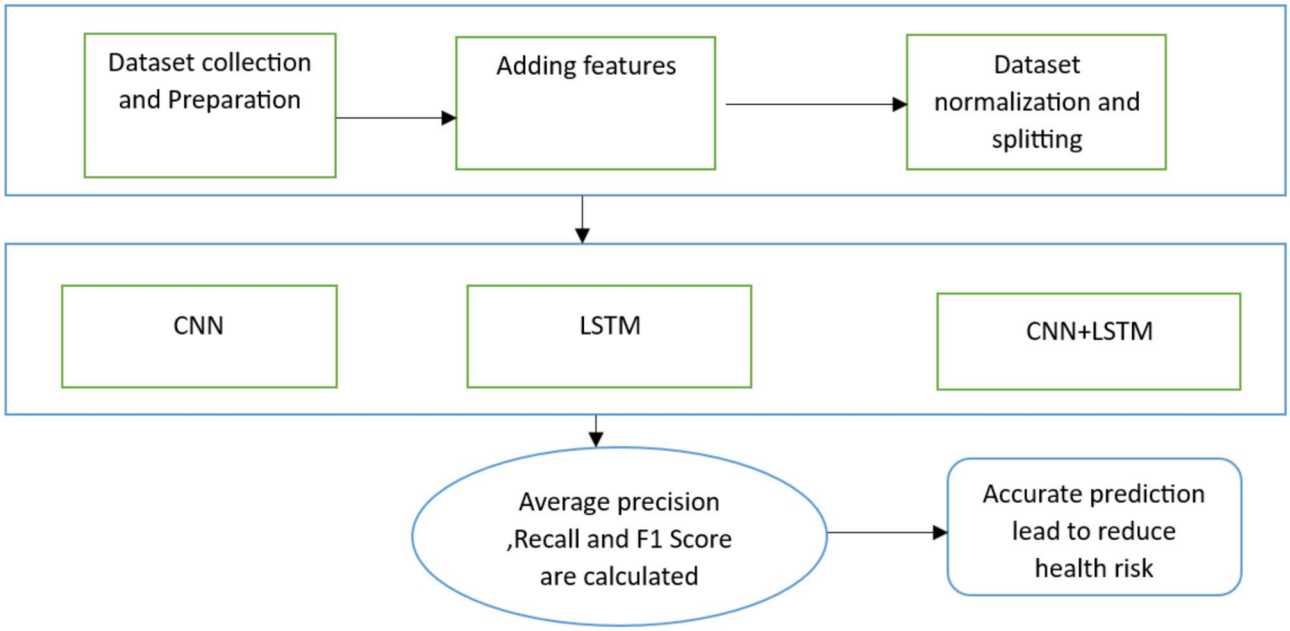

Fig. 2. Methodology for AQI Prediction and Health Risk Estimation.

maps into LSTM layers, enabling simultaneous learning of spatial representations and temporal dynamics. For model training, categorical cross-entropy is used as the loss function. The Adam optimizer is employed due to its adaptive learning rate and stable convergence properties. Model performance is evaluated using precision, recall, and F1-score, defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{32}$$

A confusion matrix is used to visualize misclassification patterns across AQI categories. In addition to prediction, the model also estimates the health risk probability based on AQI levels. After classifying the AQI, each category is mapped to a probability of disease occurrence based on public health data. For example, higher AQI levels (such as "Very Poor" or "Severe") are associated with increased risks of respiratory and cardiovascular diseases. This enables proactive healthcare responses and urban air management policies. The overall pipeline, from data acquisition to disease risk estimation, is illustrated in Fig. 2.

The hyperparameters of the proposed hybrid CNN–LSTM model were selected through systematic empirical evaluation using the validation set. Preliminary experiments with smaller network capacities resulted in underfitting, particularly in modeling abrupt AQI fluctuations arising from complex pollutant interactions. Consequently, higher-capacity configurations were adopted to effectively capture non-linear spatial features and long-term temporal dependencies. To mitigate overfitting, dropout regularization and early stopping based on validation loss were applied. The finalized hyperparameter configuration is summarized in Table 6.

Results and discussion

All models were trained using a batch size of 32 and 400 epochs with early stopping patience of 30 epochs. The dataset comprises observational and meteorological records for Gurugram city collected from 10 January to 9 April 2024 via the OpenWeather API. To mitigate overfitting, training employed EarlyStopping with a patience of 30 epochs (monitoring validation loss) and ReduceLROnPlateau to lower the learning rate when validation metrics plateaued. Each experiment was repeated five times and results reported are the averages across those independent runs. The Adam optimizer was used as the default optimization method for the experiments reported below.

Hyperparameter	Value
Input shape	(N, 1) (number of features × channels)
CNN filters	[64, 64, 64, 64]
Kernel size	5
Pooling size	2
LSTM memory units	[64, 64, 64, 64, 64]
Hidden units (Dense layer)	128
Activation function	Swish
Dropout rate	0.5
Optimizer	Adam
Loss function	Categorical Cross-Entropy
Batch size	128
Maximum epochs	400
Early stopping patience	30 epochs
Learning rate scheduler	ReduceLROnPlateau (factor = 0.5, patience = 40)
Evaluation metrics	Accuracy, Precision, Recall, F1-score

Table 6. Hyperparameter configuration of the proposed hybrid CNN–LSTM model.

Model	Presence of all features	Absence of features
CNN	0.8917	0.6901
LSTM	0.8750	0.9301
CNN+LSTM	0.9101	0.8600

Table 7. Average Precision (presence vs absence of engineered features).

Model	Presence of all features	Absence of features
CNN	0.8589	0.7237
LSTM	0.8850	0.8101
CNN+LSTM	0.9118	0.9091

Table 8. Average Recall (presence vs absence of engineered features).

Hyperparameter tuning and architecture choices

Hyperparameter searches were conducted to identify configurations that improve generalization and stability. When engineered external features (time cyclic encodings, ratios, month/date, moving averages, and lagged variables) were included, regularization was strengthened (L2 regularization and dropout set to 0.5) and larger-capacity networks were favored (e.g., 128 dense neurons, 256 convolutional filters and up to 512 LSTM memory units, four convolutional layers, and up to three stacked LSTM layers). In contrast, when features were omitted, smaller configurations (e.g., 64 filters / 64 memory units with dropout = 0.2) yielded better trade-offs between capacity and overfitting. These tuned settings served as the basis for the comparative experiments described in the following subsections.

Effect of engineered features: precision

We evaluated the influence of engineered features by comparing model precision with and without the full set of external features. The hybrid CNN–LSTM maintains the highest average precision when features are present, while LSTM shows an unexpectedly higher precision in the no-features condition; this anomaly suggests that some engineered inputs may interact suboptimally with the LSTM under the current hyperparameterization. Quantitative values for average precision are presented in Table 7 and discussed in the subsequent text. Table 7 shows that the CNN improves substantially with engineered inputs (from 0.6901 to 0.8917), the hybrid architecture attains the best precision with features (0.9101), and the LSTM achieves 0.8750 with features but 0.9301 without features.

Effect of engineered features: recall

The recall results indicate that the hybrid model is most successful at capturing true positive AQI instances across both input conditions, demonstrating robustness to the inclusion or exclusion of engineered features. The LSTM shows improved recall with features (0.8850) relative to without (0.8101), suggesting better sensitivity when temporal and engineered signals are available. Detailed recall values are listed in Table 8. As reported in

Model	Presence of all features	Absence of features
CNN	0.8674	0.7036
LSTM	0.8788	0.8357
CNN+LSTM	0.9109	0.8808

Table 9. Average F1-Score (presence vs absence of engineered features).

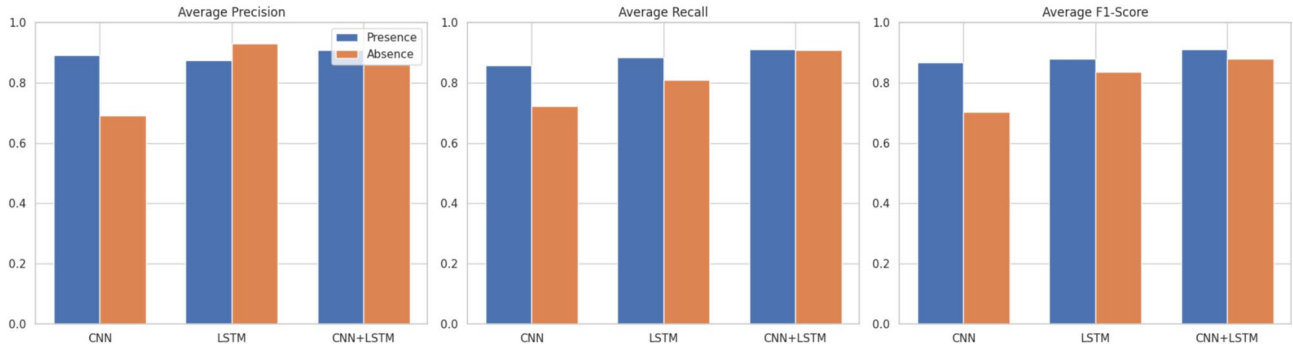

Fig. 3. Evaluation Metrics Comparison.

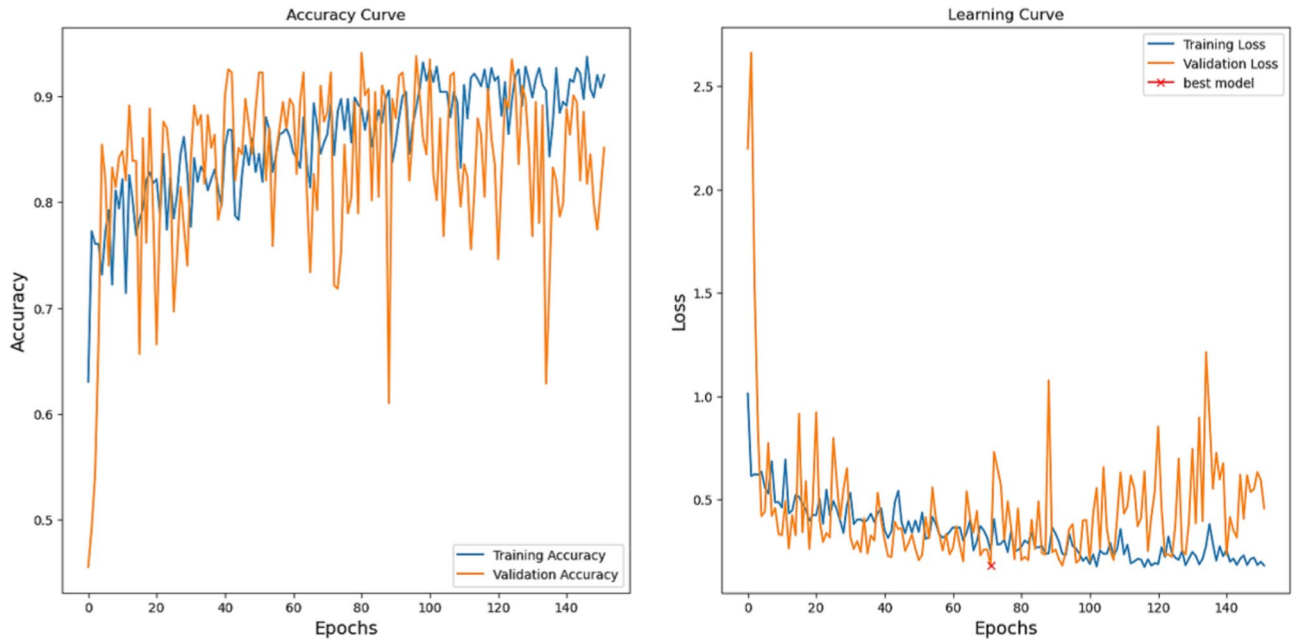

Fig. 4. Loss and Accuracy per epoch Curve of Hybrid CNN-LSTM Model.

Table 8, the CNN recall improves markedly with features (0.8589 vs 0.7237) and the hybrid achieves the highest recall in both scenarios (0.9118 presence, 0.9091 absence).

Effect of engineered features: F1-score

The F1-score, which balances precision and recall, confirms that the hybrid CNN–LSTM attains the best overall trade-off when engineered features are included. The CNN benefits greatly from engineered inputs, moving from a modest F1 of 0.7036 (no features) to 0.8674 (with features). The LSTM also improves F1 when features are used (0.8788 vs 0.8357). Table 9 summarizes these F1 comparisons and supports the conclusion that combining convolutional feature extraction with recurrent temporal modeling is the most effective approach for this AQI prediction task under the tested configurations. Figure 3 presents the Evaluation Metric Comparison of CNN, LSTM with hybrid CNN-LSTM Model. Figure 4 shows the learning curve which decreases loss on testing dataset.

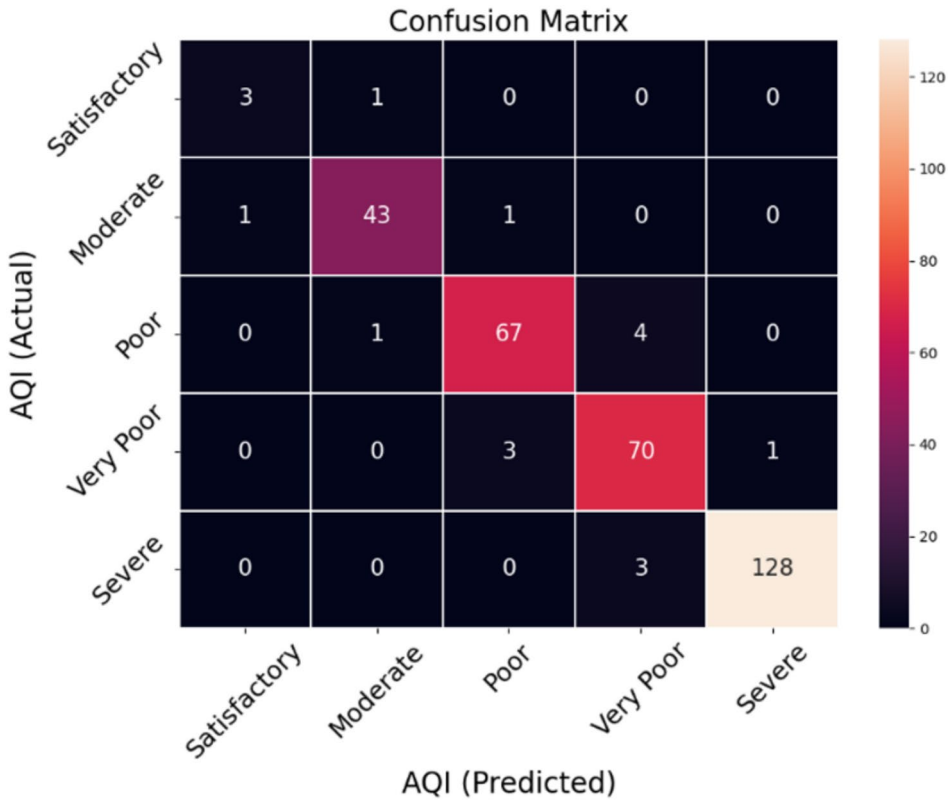

Fig. 5. Confusion Matrix of Hybrid CNN-LSTM Model.

Comparison	Metric	*p*-value
CNN vs CNN+LSTM	F1-score	0.001953
LSTM vs CNN+LSTM	F1-score	0.001953
CNN vs LSTM	F1-score	0.921875

Table 10. Statistical significance analysis under presence of engineered features using Wilcoxon signed-rank test.

The class-wise performance of the proposed Hybrid CNN-LSTM model is further analyzed using the confusion matrix shown in Fig. 5 to complement the aggregate metrics reported. The model achieves strong recall for critical AQI categories, with recall values of 0.977 for *Severe*, 0.946 for *Very Poor*, 0.931 for *Poor*, 0.956 for *Moderate*, and 0.750 for *Satisfactory*. Similarly, high precision is observed for extreme pollution levels, particularly for the *Severe* class (0.992) and the *Poor* class (0.944), indicating reliable identification of hazardous air quality conditions. Minor misclassification is primarily observed between adjacent AQI categories, such as *Poor* and *Very Poor*, due to overlapping pollutant concentration ranges and gradual transitions in air quality levels. Importantly, severe errors across distant categories (e.g., *Satisfactory* misclassified as *Severe*) are negligible, demonstrating that the model preserves the ordinal structure of AQI levels. These results confirm that the class-wise precision, recall, and F1-scores summarized are supported by consistent per-class behavior, particularly for high-risk AQI categories that are critical for public health decision-making.

Statistical Test

To statistically validate the observed performance differences under the presence of engineered features, a Wilcoxon signed-rank test was conducted to compare CNN, LSTM, and the proposed hybrid CNN–LSTM model. As shown in Table 10, the hybrid CNN–LSTM model achieves statistically significant improvements over both CNN and LSTM in terms of F1-score ($p = 0.001953$), confirming that its superior performance is not due to random variation. In contrast, the difference between CNN and LSTM is not statistically significant ($p = 0.921875$), indicating comparable predictive capability when used independently. These results reinforce the effectiveness of the proposed hybrid architecture in leveraging complementary spatial and temporal representations when all engineered features are present.

To assess the robustness of the evaluated models, we further analyze performance variability across repeated runs under the presence of engineered features. Each model was trained and evaluated over multiple independent

Model	Precision	Recall	F1-score
CNN	0.8917 ± 0.010	0.8589 ± 0.012	0.8674 ± 0.011
LSTM	0.8750 ± 0.009	0.8850 ± 0.010	0.8788 ± 0.009
CNN–LSTM	$\mathbf{0.9101 \pm 0.008}$	$\mathbf{0.9118 \pm 0.007}$	$\mathbf{0.9109 \pm 0.006}$

Table 11. Performance comparison with mean ± standard deviation across repeated runs (presence of engineered features).

Model	Adam	RMSprop
CNN	0.8674	0.6190
LSTM	0.8788	0.6206
CNN+LSTM	0.9250	0.9041

Table 12. Average F1-Score using Adam versus RMSprop.

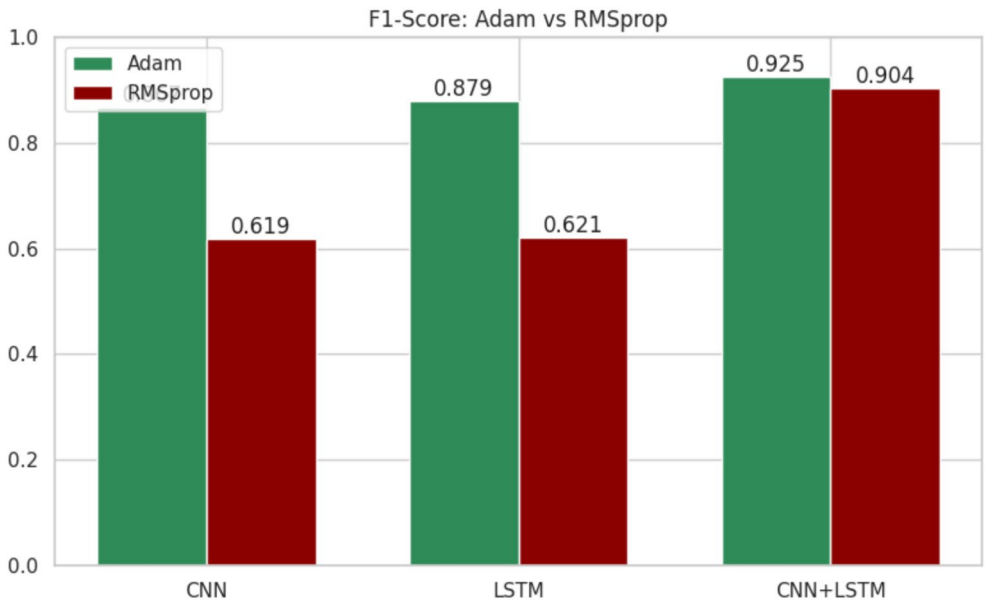

Fig. 6. Evaluation Comparison for Different Optimizers.

executions with different random seeds, and the mean and standard deviation of precision, recall, and F1-score were computed. Reporting variability alongside average values provides a more reliable indication of model stability and reduces the risk of overinterpreting marginal improvements.

As shown in Table 11,the proposed hybrid CNN–LSTM model not only achieves the highest average precision, recall, and F1-score under the presence of engineered features, but also exhibits lower standard deviation across repeated runs, indicating improved stability. While the numerical gains over individual CNN and LSTM models are moderate, the reduced variance and consistent performance across runs support the robustness of the hybrid architecture. These results complement the statistical significance analysis and reinforce that the observed improvements are systematic rather than incidental.

Optimizer comparison

To determine the effect of optimizer choice on final performance, we repeated experiments using RMSprop and compared results with the default Adam runs. The comparative F1-scores are given in Table 12 and Fig. 6. As shown in Table 12, Adam consistently produced higher F1-scores for CNN and LSTM in our setup, and it provided a modest advantage for the hybrid architecture as well. These findings indicate that Adam's combination of adaptive learning rates and momentum is beneficial for the models and data used in this study.

Disease risk probability

Table 13 reports the estimated disease risk probabilities produced by each model for Gurugram city. These probabilities were derived from the models' predicted air-quality outcomes and then mapped to a simple risk index as described in the Methodology; the table presents the average risk probability returned by each

Sno	Algorithm	Disease Risk Probability
1	CNN	46.33%
2	LSTM	47.36%
3	Hybrid	49.43%

Table 13. Disease risk probability.

AQI category	Relative risk score	Health interpretation
Good	0.05	Minimal risk; air quality poses little or no health concern.
Satisfactory	0.10	Low risk; acceptable air quality for most individuals.
Moderate	0.25	Mild risk; may cause minor discomfort to sensitive groups.
Poor	0.50	Elevated risk; increased likelihood of respiratory symptoms.
Very Poor	0.75	High risk; prolonged exposure may aggravate existing diseases.
Severe	0.90	Very high risk; significant health impacts likely for all groups.

Table 14. Mapping of AQI categories to relative health risk levels.

algorithm across the evaluation runs. Table 13 shows that the hybrid CNN–LSTM model yields the highest estimated disease risk (49.43%), while the CNN produces the lowest estimate (46.33%), with the LSTM falling in between (47.36%).

The differences in estimated risk are modest but relevant for public-health planning: a higher percentage indicates that, under the model's mapping from predicted air-quality levels to health risk, a larger portion of the population is expected to be exposed to conditions associated with increased disease probability. The hybrid model's higher value suggests that its combined feature-extraction and temporal-modeling capacity identifies patterns that translate to slightly greater projected health burden than either standalone CNN or LSTM in this dataset. However, these model-derived probabilities should be interpreted cautiously. They depend on the exposure-to-risk mapping used, the representativeness of the underlying AQI predictions, and assumptions made in the risk-calculation step. We therefore recommend treating these results as relative indicators (comparative model outputs) rather than absolute epidemiological estimates. For rigorous use in policy or clinical contexts, follow-up steps are necessary: (1) perform statistical testing (e.g., paired comparisons on per-run estimates) to determine whether the observed differences are significant; (2) validate the risk mapping against local health or clinical data where available; and (3) conduct sensitivity analyses to quantify how changes in AQI thresholds or mapping rules affect the estimated probabilities. These actions will help convert the model outputs in Table 13 into more actionable and trustworthy guidance for stakeholders.

Health risk estimation based on AQI predictions

To enhance interpretability of AQI predictions, a health risk estimation component is incorporated to translate predicted AQI categories into relative disease risk indicators. The mapping between AQI categories and health risk scores is explicitly defined in Table 14. This mapping is derived from qualitative health advisory guidelines issued by environmental and public health agencies, which associate worsening air quality levels with progressively higher respiratory and cardiovascular risk. Each AQI category is assigned a normalized relative risk score between 0 and 1, reflecting the monotonic increase in health vulnerability as air quality deteriorates. Lower AQI categories such as *Good* and *Satisfactory* correspond to minimal risk, whereas higher categories such as *Very Poor* and *Severe* indicate substantially elevated risk due to sustained exposure to high concentrations of pollutants including $PM_{2.5}$, PM_{10}, NO_2, and SO_2. The assigned risk scores are heuristic and intentionally coarse-grained, designed to preserve ordinal severity rather than quantify absolute disease incidence. It is important to emphasize that the estimated health risk values represent **relative indicators only**. They do not constitute epidemiological predictions, clinical diagnoses, or individualized health assessments. The proposed framework does not model exposure duration, demographic vulnerability, or pre-existing medical conditions, and therefore should be interpreted solely as a supportive awareness tool. Ethical considerations are explicitly acknowledged, and the health risk estimation component is intended to complement established public health advisories rather than replace expert medical or policy guidance.

Discussion

Across the conducted experiments, the hybrid CNN–LSTM architecture consistently provides a balanced trade-off between precision and recall when engineered temporal features are available. The CNN benefits substantially from feature engineering, as handcrafted temporal descriptors enhance its ability to extract discriminative local patterns. In contrast, the standalone LSTM exhibits comparatively strong performance even in the absence of engineered features, and in some cases achieves higher precision. This behavior can be attributed to the LSTM's inherent capability to directly model sequential dependencies from raw pollutant time series, where a reduced and less redundant feature space may lead to more conservative predictions and fewer false positives. When engineered features are introduced, the increased feature complexity may not uniformly benefit single-branch temporal models and can occasionally introduce redundancy, explaining the observed reduction in LSTM

precision. The hybrid CNN–LSTM model mitigates this issue by jointly learning complementary spatial feature representations and long-term temporal dependencies, resulting in more stable performance across metrics. Class-wise analysis and the confusion matrix (Fig. 5) further reveal that most misclassifications occur between adjacent AQI categories (e.g., *Poor–Very Poor* and *Very Poor–Severe*), while extreme category confusions are rare. Notably, the hybrid model improves recall for high-risk categories such as *Severe*, which is particularly important from a public health perspective. Model performance is expected to vary across seasons, cities, and pollution regimes due to differences in emission sources, meteorological conditions, and pollutant dispersion patterns. For example, winter seasons characterized by temperature inversion and reduced atmospheric mixing may amplify the importance of lagged and moving-average features, while summer or monsoon periods may introduce higher variability due to wind and precipitation effects. Similarly, cities dominated by traffic-related emissions may benefit more from pollutant ratio features (e.g., CO/NO_2), whereas industrial regions may exhibit different dominant patterns. As a result, while the proposed framework is methodologically generalizable, model performance and optimal feature configurations may differ across regions and seasons. Overall, although the numerical performance gains of the hybrid model over individual architectures are sometimes modest, they are consistent and accompanied by improved robustness across AQI classes and pollution regimes. Therefore, the hybrid CNN–LSTM architecture is recommended for operational AQI forecasting when computational resources permit. In resource-constrained settings, a carefully tuned LSTM remains a viable alternative. In all cases, appropriate hyperparameter tuning, region-specific feature selection, and seasonal recalibration are essential for achieving reliable and interpretable performance.

Limitations

One important limitation of this study is the relatively short temporal span of the dataset, which covers air quality observations from 10 January to 9 May 2024. This period was selected primarily due to data availability constraints and to ensure consistency and completeness of high-resolution pollutant measurements obtained from the OpenWeather API. While this window captures meaningful short-term temporal dynamics and transition effects from winter to early summer, it does not fully encompass long-term seasonal variability such as monsoon-related pollution patterns or post-monsoon agricultural burning events. Consequently, the model's ability to generalize across all seasonal regimes may be limited. The restricted time horizon implies that certain seasonal phenomena—such as prolonged humidity-driven pollutant dispersion or extreme winter inversion effects—are underrepresented. As a result, the reported performance should be interpreted as representative of short- to mid-term AQI forecasting rather than year-round prediction. Future work will address this limitation by incorporating multi-year datasets spanning all major seasons, enabling more comprehensive seasonal modeling and improved generalization. In addition, some counterintuitive behaviors—such as higher LSTM precision in the absence of engineered features—motivate further investigation. Planned extensions include targeted ablation studies, detailed class-wise confusion analysis, feature correlation assessment, and dedicated hyperparameter sweeps for each model–feature combination.

Conclusion and future work

The experimental results indicate that, within the scope of this study, the hybrid CNN–LSTM architecture achieves the strongest overall performance for short-term AQI forecasting on the evaluated dataset, attaining an F1-score of approximately 91% compared with 87.9% for LSTM and 86.7% for CNN under the tested configurations. The results further show that combining convolutional feature extraction with recurrent temporal modeling yields more stable and consistent predictions than standalone architectures, and that the Adam optimizer provides more reliable convergence than RMSprop in this setting. It is important to emphasize that these findings are derived from a region-specific dataset collected from a single city over a limited time span, and therefore should be interpreted as a case-study evaluation rather than evidence of universal generalization. While the predicted AQI values are further mapped to relative health risk indicators to enhance interpretability, these estimates are intended solely as supportive, high-level indicators and should not be construed as direct measures of population-wide health outcomes or clinical risk. Future work will focus on extending the evaluation to multi-city and multi-season datasets to assess geographic and temporal generalizability. Additional directions include systematic hyperparameter optimization, uncertainty quantification through ensemble learning or Monte Carlo dropout, and validation of model-derived risk indicators against aggregated public health statistics. Further integration of heterogeneous data sources (e.g., satellite observations, traffic density, and land-use information), along with model compression and explainability techniques, will also be explored to support reliable and interpretable deployment in operational air-quality monitoring and early-warning systems.

Data availability

Data will be made available on request.

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Author contributions

All authors contributed equally to the conception, methodology, experimentation, analysis, manuscript preparation, and final approval of the paper.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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